Task Title

Create and Clean Application Log Data Using Python

Task Description

Create a custom CSV file that simulates raw application logs. Then clean the CSV data by removing duplicates, handling missing values, and standardizing column names.

Step 1: Create Custom CSV (Raw Logs)

```
import pandas as pd


# Create raw log data (with duplicates and missing values)
data = {
    "User ID": [101, 102, 103, 103, 104],
    "User Name": ["alice", "bob", "charlie", "charlie", None],
    "Login Time": [
        "2026-02-01 09:10",
        "2026-02-01 09:30",
        "2026-02-01 10:00",
        "2026-02-01 10:00",
        "2026-02-01 10:15"
    ],
    "Response Time (ms)": [120, 95, None, None, 130],
    "Status": ["success", "success", "failed", "failed", "success"]
}
```

```
# Create DataFrame
df_raw = pd.DataFrame(data)

# Save raw logs
df_raw.to_csv("raw_logs.csv", index=False)

print("Raw CSV logs created.")
```

Step 1 Output: Raw CSV (raw_logs.csv)

User ID	User Name	Login Time	Response Time (ms)	Status
101	alice	2026-02-01 09:10	120	success
102	bob	2026-02-01 09:30	95	success
103	charlie	2026-02-01 10:00	NaN	failed
103	charlie	2026-02-01 10:00	NaN	failed
104	NaN	2026-02-01 10:15	130	success

Step 2: Clean the Raw CSV Logs

```
import pandas as pd

# Load raw CSV
df = pd.read_csv("raw_logs.csv")

# Standardize column names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_").str.replace("(", "").str.replace(")", "")
```

```
# Remove duplicate rows
df = df.drop_duplicates()

# Handle missing values
df["user_name"] = df["user_name"].fillna("unknown")
df["response_time_ms"]=df["response_time_ms"].fillna(df["response_time_ms"].mean())

# Save cleaned CSV
df.to_csv("cleaned_logs.csv", index=False)
print("Cleaned CSV logs saved successfully.")
```

Step 2 Output: Cleaned CSV (cleaned_logs.csv)

user_id	user_name	login_time	response_time_ms	status
101	alice	2026-02-01 09:10	120.0	success
102	bob	2026-02-01 09:30	95.0	success
103	charlie	2026-02-01 10:00	115.0	failed
104	unknown	2026-02-01 10:15	130.0	success

Step	Action
Column names	Converted to lowercase & standardized
Duplicates	Removed
Missing names	Filled with "unknown"
Missing response time	Filled with mean value
Final file	cleaned_logs.csv

Data Cleaning Operations Performed

1. Column Name Standardization

- Removed extra spaces, brackets, and special characters
- Converted all column names to lowercase
- Replaced spaces with underscores

Why?
Ensures consistency, avoids errors in coding, and makes columns easier to access programmatically.

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2. Duplicate Record Removal

- Identified and removed repeated rows from the dataset

Why?
Duplicates can bias analysis results and negatively affect machine learning model performance.

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3. Handling Missing Categorical Values

- Replaced missing values in the user_name column with a default value ("unknown")

Why?
Prevents data loss and allows categorical fields to remain usable for analysis and modeling.

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4. Handling Missing Numerical Values

- Filled missing values in the response_time_ms column using the **mean value**

Why?
Maintains numerical consistency and avoids dropping rows that contain valuable information.

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5. Data Type Preservation

- Ensured numerical columns remain numeric and categorical columns remain textual

Why?
Correct data types are essential for calculations, visualizations, and ML algorithms.

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6. Schema Consistency

- Created a uniform and clean structure across all records

Why?
A consistent schema ensures the dataset is ready for:

- Exploratory Data Analysis (EDA)
- Machine Learning
- Reporting and visualization